TED UNIVERSITY

Department of Computer Engineering

CMPE 491 – Capstone Design Project I | 2026–2027

PerDrobe Project Proposal

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Project name: PerDrobe **Project webpage:** https://perdrobe2026-commits.github.io/

Team Members

Name	Student number
Yiğit Özmut	10039204492
Atay Aktürk	19223226142
Seyid Berat Ergün	10012213640

Academic Supervision

Supervisor Ayşenur Birtürk

Jury Member 1 Mehmet Evren Coşkun

Jury Member 2 Eren Ulu

Course Coordinator Prof. Dr. Gökçe NUR YILMAZ

Project Description

PerDrobe is a ai integrated personalized wardrobe and clothing recommendation app we plan to develop for our CMPE 491–492 project. We want to make it easier for people to create outfits using clothes they already own, especially when they find a style they like but are unsure how to recreate it. Users will be able to build a private digital wardrobe by uploading photos of their own clothes, and the system will help organize these items by removing the background and focusing on the clothing piece itself. This will allow wardrobe items to be stored in a cleaner and more usable way. Users can also enter their preferred styles and fits, while their likes, saved outfits, and followed accounts will help the system better understand their preferences. In addition to recommending individual items, the app will allow users to combine clothing pieces and create outfits from their wardrobe. With Outfit Remix, users will be able to recreate outfits they discover by replacing some pieces with suitable alternatives from their own clothes. The app will also suggest new clothing items that match what users already have. To support shopping decisions, we plan to provide size suggestions using optional body measurements, usual sizes, and available product size charts, but only when there is enough information to make a reasonable recommendation.

The main technical part of our project is an AIbased recommendation system that identifies suitable clothing items, analyzes how well they work together, and ranks them based on each user's preferences. Another important part of the system will be processing uploaded clothing images so that user-owned items can be cleaned, represented digitally, and used in outfit generation. We plan to compare three approaches. The first will use clothing attributes, the second will also consider user preferences and interaction signals, and the third will additionally include the user's existing wardrobe. We will evaluate how relevant and useful the recommendations are, how much variety they provide, how well they cover the available items, and how quickly the system responds. Size and fit recommendations will be evaluated only when suitable reference data is available.